# Francesco Sorrenti

Postdoctoral Researcher

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https://fsorrenti.github.io/
ORCID | GitHub
LinkedIn


## Research Interests

- Weak lensing and magnification bias (*Euclid*)
- LSS modelling
- Type Ia supernovae, peculiar velocities and Local structure
- Cosmic dipoles and isotropy of the Universe
- Consistency tests of the Standard Model and $H_0$ systematics

## Numerical Skills

**Programming:**
Python, JAX, Mathematica, LaTeX, C++

**OS & Tools:**
HPC, Git, macOS, Windows, Ubuntu

## Software

- **scoutpip**: Lead developer
- **coffe**: Contributor

## Languages

**Italian:** native
**English:** C1
**French:** B1
**Spanish:** B1
**Catalan:** A2

## Current & Previous Positions

Oct 2025 – Pres. — **SNSF Postdoc.Mobility Fellowship** — ICCUB
ICC, University of Barcelona, Spain.

Jul–Sep 2025 — **Postdoctoral Researcher** — Uni. Geneva, Switzerland
Department of Theoretical Physics.

## Education

2021 – 2025 — **PhD in Cosmology** — Uni. Geneva, Switzerland
Supervisors: Profs. R. Durrer, M. Kunz. Funded by SNSF Sinergia.

2018 – 2021 — **Master in Physics** — Uni. Perugia, Italy
Final Grade: 110/110. Supervisor: Prof. M. Gerbino.

2015 – 2018 — **Bachelor in Physics** — Uni. Perugia, Italy
Final Grade: 110/110 cum laude. Supervisor: Prof. S. Pacetti.

## Memberships & Academic Service

2026 – Pres. — **Journal referee for JCAP**

2023 – Pres. — **COST Action** — CA21136
Member of the CosmoVerse network.

2021 – Pres. — **Euclid Consortium**
- Member of the Euclid Consortium Diversity Committee.
- Chair, then Co-chair, of the Early Career Committee (2023–2026).
- Coordinator of the DR1-KP-TH2 Copernican Principle project.
- Theory Working Group (Consistency effects WP) and DR1-KP-GC8 (working on Magnification bias).

2019 – 2022 — **SIF & AISF** — Member
Italian Physical Society; Italian Association of Physics Students.

## Selected Publications (full list on Google Scholar and INSPIRE)

2026 — **Preprint** — JCAP (submitted)
*Reconstructing the kinematics of Laniakea using Type Ia Supernovae.* **F. Sorrenti**, E. Pastén, L. Giani. 2607.11860

2024 — **Journal article** — A&A
*Euclid preparation XL. Impact of magnification on spectroscopic galaxy clustering.* G. Jelic-Cizmek, **F. Sorrenti**, et al. 2311.03168

2023 — **Journal article** — JCAP
*The Dipole of the Pantheon+SH0ES Data.* **F. Sorrenti**, R. Durrer, M. Kunz. 2212.10328

Oct 2026 — **Bibliometric Summary**
- Scopus: 759 citations, $h$-index 5
- Web of Science: 668 citations, $h$-index 5

## Fellowships, Prizes & Funding

2025 – 2027 — **SNSF Postdoc. Mobility Fellowship** — CHF 103,700
Postdoctoral fellowship.

Apr 2025 — **SNSF Funding** — CHF 2,500
As organiser: CosmoFONDUE 2025 conference.

Feb 2025 — **Sacad Funding** — CHF 10,000
As organiser: CosmoFONDUE 2025 conference.

Feb 2025 — **Schmidheiny Foundation Scholarship** — CHF 1,812
Funding for Cosmic Flows 2025 in Brisbane, Australia.

May 2024 — **ESA Scholarship** — CHF 35,000
Team prize, "Space and Global health" hackathon.

Feb 2024 — **CosmoVerse Scholarship** — EUR 2,000
Funding for a visit to the University of Turin.

Nov 2023 — **Sacad Scholarship** — CHF 2,300
Funding for a visit to the University of Oxford.

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## Presentations (here for the full list)

| | | |
|---|---|---|
| Sep 2026 | **Cosmology from Home 2026**<br>Invited panelist. | Online |
| Sep 2026 | **Astrophysics and Space Science III**<br>Talk. | Jesi, Italy |
| Jul 2026 | **Exploring New Frontiers in Cosmology**<br>Invited talk (Workshop) and Short Talk (Conference). | GGI Florence, Italy |
| Jun 2026 | **LUPM Seminar**<br>Seminar. | Montpellier, France |

## Teaching & Supervising

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|---|---|---|
| 2026 | **Master's Thesis Supervisor**<br>ICC. Project on SNe Ia lensing impact. | Uni. Barcelona, Spain |
| 2021 – 2025 | **Teaching Assistant**<br>• *Mechanics II* (Bachelor's) — 6 CFU/sem., 52h (2 sem.)<br>• *Theoretical Lab. IV* (Master's) — 15 CFU, 52h/yr (2 yrs)<br>• *Group Theory* (Master's) — 7 CFU, 52h (1 sem.)<br>• *Quantum Mechanics II* (Bachelor's) — 6 CFU (1 sem.) | Uni. Geneva, Switzerland |
| 2020 – 2021 | **Academic Tutor**<br>• *General Physics* (Eng. Bachelor's) — 12 CFU, 100h (1 yr)<br>• *Physics* (Env. Sciences Bachelor's) — 6 CFU, 50h/sem. (2 sem.) | Uni. Perugia, Italy |

## Visits & Conference Organisation

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|---|---|---|
| Jun 2025 | **Conference Organiser**<br>"CosmoFONDUE" conference (next edition: Jan 2027). | CERN / Uni. Geneva, Switzerland |
| Mar 2025 | **Visiting Researcher**<br>Cosmology Group (Prof. Luongo). | Uni. Camerino, Italy |
| Jun 2024 | **SOC Member**<br>Euclid Consortium meeting. | Rome, Italy |
| Jan 2024 | **Visiting Researcher**<br>Cosmology Group (Prof. Camera). | Uni. Turin, Italy |
| Nov–Dec 2023 | **Visiting Researcher**<br>Beecroft Institute for Particle Astro. & Cosmo. (Prof. Ferreira). | Uni. Oxford, UK |
| 2017 – 2023 | **Schools**<br>*Cargèse, France* (2023), *Les Houches, France* (2022), *GGI Florence, Italy* (2022), *GIPE Dresden, Germany* (2019), *NiPS Gubbio, Italy* (2017). | |

## Outreach Activities

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|---|---|---|
| 2023 – 2024 | **Euclid Swiss Outreach**<br>Contributed to the promotion of the Euclid mission. | Switzerland |
| 2019 – 2021 | **AISF Perugia**<br>Talks with M. Gerbino (INFN) and V. Vagelli (ASI); social media and web. | Event Organisation, Italy |
| 2019 – 2021 | **Science outreach**<br>Workshops for children, theatre-based outreach, student orientation. | Uni. Perugia & POST, Italy |